To: The Editorial Board, Scientific Reports (Nature Portfolio)

From: Mr. Ranjith Raj Anbalagan, Research Scholar,

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Dear Editor,

We are pleased to submit our manuscript entitled "Adaptive Multi-Objective Chaotic S-Box Design for Lightweight and Secure IoT Communication" for consideration in Scientific Reports. The work is a four-author collaboration between R. Saktheeswari (Sri Venkateswara College of Engineering, India), Ranjith Raj Anbalagan (NIT Tiruchirappalli, India), Aravind Pitchai Venkataraman (St. Joseph University in Tanzania), and Kehali Anteneh Jember (Adama Science & Technology University, Ethiopia; corresponding author).

We couple a Non-dominated Sorting Genetic Algorithm II (NSGA-II) with the Flower Pollination Algorithm (FPA) over a two-dimensional Logistic map and expose a full four-objective Pareto front in the (NL_min, LE, DU, runtime) space. The selected knee-point S-box reaches a nonlinearity of 102, SAC of 0.503, differential uniformity of 10, NPCR of 99.51 percent, and UACI of 33.57 percent at an estimated lookup cost of 0.83 ms per byte on an ARM Cortex-M0+ profile.

A five-seed ablation makes clear that the headline contribution is the multi-objective framework itself rather than the FPA perturbation, which acts as an archive-diversity tie-breaker but does not dominate NSGA-II on any single metric. The manuscript is within Scientific Reports' scope (computer science / cryptography / IoT security), respects all stated limits (title 11 words, abstract 191 words, references 29, display items 8, page count 13), declares no competing interests, and is not under consideration elsewhere. All numerical results come from a deterministic Python simulator (seed 20260814) released with the manuscript for full

reproducibility. Hardware measurement on representative Cortex-M boards is the next planned step.

Thank you for considering our manuscript. We look forward to your response.

Sincerely,

Mr. Ranjith Raj Anbalagan, Research Scholar,

(on behalf of all co-authors)